Year 11 Mathematics

Algebra 9

7.1 Formulae to Solve Proportion Problems

Formulae to Solve Proportion Problems

STARTER · 5 MIN

Date: _____ **Title:** Formulae to Solve Proportion Problems

LEARNING OBJECTIVES

- Recognise direct proportion and use $y = kx$
- Use proportion with squares, cubes and square roots
- Solve problems involving direct proportion

STARTER *Answer in your book. No calculators.*

1. 3 pens cost £1.20. Find the cost of 5 pens.
2. Solve $12 = 4k$
3. Solve $50 = 2k$
4. Work out 4×3^2

Starter — Answers

STARTER

Tick or correct in green pen

1. $1.20 \div 3 = 0.40, \quad 5 \times 0.40 = \mathbf{£2}$

2. $k = \mathbf{3}$

3. $k = \mathbf{25}$

4. $4 \times 9 = \mathbf{36}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can recognise direct proportion and use y = kx	☐
SILVER	I can use proportion with squares, cubes and square roots	☐
GOLD	I can solve problems involving direct proportion	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Direct Proportion

INTRODUCTION · 3 MIN

If y is directly proportional to x, then as x doubles, y doubles. Write it as a formula with a constant k.

The symbol ∝

∝ means 'is proportional to'.

$y \propto x$

Formula

Replace ∝ with = k.

$y = kx$

Find k

Substitute a known pair of values.

$12 = 4k \rightarrow k = 3$

Other powers

Use the power in the question.

$y \propto x^2 \rightarrow y = kx^2$

Examples 1 and 2 — Finding the Formula

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint $y \propto x \rightarrow y = kx$

Example 1

y is directly proportional to x. y = 15 when x = 3. Find a formula for y in terms of x.

Example 2

y = 5x. Find y when x = 8, and x when y = 60.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$$y = kx$$
$$15 = 3k$$
$$k = 5$$
$$\mathbf{y = 5x}$$

Tip: find k first.

Example 2

$$y = 5 \times 8$$
$$= 40$$
$$60 = 5x$$
$$\mathbf{x = 12}$$

Tip: substitute, then solve.

Task 1

PRACTICE · 9 MIN

BRONZE *y is directly proportional to x*

Hint $y \propto x \rightarrow y = kx$

1. $y = 20$ when $x = 4$. Find k.
2. $y = 18$ when $x = 6$. Find a formula.
3. Use $y = 3x$ to find y when $x = 7$.
4. $d \propto t$. $d = 150$ when $t = 3$. Find d when $t = 5$.
5. $C \propto n$. $C = 24$ when $n = 8$. Find n when $C = 45$.

★ $y = 7$ when $x = 2$. Find y when $x = 9$.

Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1. $20 = 4k, \quad k = \mathbf{5}$

2. $18 = 6k, \quad y = \mathbf{3x}$

3. $y = 3 \times 7 = \mathbf{21}$

4. $k = 50, \quad d = 50 \times 5 = \mathbf{250}$

5. $k = 3, \quad 45 = 3n, \quad n = \mathbf{15}$

★ $k = 3.5, \quad y = 3.5 \times 9 = \mathbf{31.5}$

Examples 3 and 4 — Squares, Cubes and Roots

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $y \propto x^2 \rightarrow y = kx^2$

Example 3

$y \propto x^2$. $y = 48$ when $x = 4$.

Find y when $x = 5$.

Example 4

$y \propto \sqrt{x}$. $y = 12$ when $x = 9$.

Find y when $x = 25$.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$y = kx^2$$
$$48 = 16k$$
$$k = 3$$
$$y = 3 \times 25$$
$$\mathbf{= 75}$$

Tip: square x before substituting.

Example 4

$$y = k\sqrt{x}$$
$$12 = 3k$$
$$k = 4$$
$$y = 4 \times 5$$
$$\mathbf{= 20}$$

Tip: $\sqrt{9} = 3$.

Task 2

PRACTICE · 9 MIN

SILVER *Answer in your book*

Hint $y \propto x^2 \rightarrow y = kx^2$

1. $y \propto x^2$. $y = 50$ when $x = 5$. Find y when $x = 3$.

2. $y \propto x^3$. $y = 16$ when $x = 2$. Find y when $x = 3$.

3. $y \propto \sqrt{x}$. $y = 10$ when $x = 4$. Find y when $x = 49$.

4. $E \propto v^2$. $E = 90$ when $v = 3$. Find E when $v = 5$.

5. $y \propto x^2$. $y = 36$ when $x = 3$. Find x when $y = 100$ $(x > 0)$.

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $k = 2, \quad y = 2 \times 9 = \mathbf{18}$

2. $k = 2, \quad y = 2 \times 27 = \mathbf{54}$

3. $k = 5, \quad y = 5 \times 7 = \mathbf{35}$

4. $k = 10, \quad E = 10 \times 25 = \mathbf{250}$

5. $k = 4, \quad x^2 = 25, \quad x = \mathbf{5}$

Examples 5 and 6 — Proportion Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $y \propto x^3 \rightarrow y = kx^3$

Example 5

(3 marks)

The distance d a ball falls is proportional to the square of the time t. It falls 20 m in 2 s. How far does it fall in 5 s?

Example 6

(2 marks)

$y \propto x^2$. What happens to y when x is doubled?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$d = kt^2$

$20 = 4k$

$k = 5$

$d = 5 \times 25$

$= 125$ m

Tip: write the formula first.

Example 6

$y = k(2x)^2$

$y = 4kx^2$

y is multiplied by 4

Tip: replace x with 2x.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $y \propto x^3 \rightarrow y = kx^3$

1. The mass m of a sphere is proportional to the cube of its radius r. m = 40 g when r = 2 cm. Find m when r = 3 cm. (3 marks)

2. The cost C of a square rug is proportional to the square of its length. A 2 m rug costs £60. Find the cost of a 3 m rug. (3 marks)

3. $y \propto x^3$. What happens to y when x is doubled? (2 marks)

4. $T \propto \sqrt{L}$. T = 2 when L = 1. Find L when T = 6. (3 marks)

5. P is proportional to the cube of v. P = 250 when v = 5. Find v when P = 16. (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $k = 5, \quad m = 5 \times 27 = \mathbf{135\ g}$

2. $k = 15, \quad C = 15 \times 9 = \mathbf{£135}$

3. $2^3 = 8$, **y is multiplied by 8**

4. $k = 2, \quad 6 = 2\sqrt{L}, \quad L = \mathbf{9}$

5. $k = 2, \quad v^3 = 8, \quad v = \mathbf{2}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE $y \propto x$. $y = 30$ when $x = 5$. Find k.

SILVER $y \propto x^2$. $y = 12$ when $x = 2$. Find y when $x = 4$.

GOLD $y \propto x^2$. What happens to y when x is tripled?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $30 = 5k, \quad k = \mathbf{6}$

SILVER $k = 3, \quad y = 3 \times 16 = \mathbf{48}$

GOLD $3^2 = 9$, **y is multiplied by 9**

Exam Practice — Direct Proportion

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $y \propto x \rightarrow y = kx$

Question 1

(3 marks)

y is directly proportional to x.

y = 28 when x = 4.

(a) Find a formula for y in terms of x. (2)

.....

(b) Find y when x = 9. (1)

y = ...

Question 2

(3 marks)

The cost, C, of some rope is directly proportional to its length, L.

6 m of rope costs £9.

Find the cost of 14 m of rope.

..... £

Exam Practice — Squares, Cubes and Roots

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $y \propto x^2 \rightarrow y = kx^2$

Question 3

(3 marks)

y is directly proportional to x^2.

y = 75 when x = 5.

Find y when x = 8.

y = ...

Question 4

(3 marks)

y is directly proportional to the square root of x.

y = 12 when x = 9.

Find y when x = 64.

y = ...

Exam Practice — Proportion Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $y \propto x^3 \rightarrow y = kx^3$

Question 5

(4 marks)

The mass, m grams, of a sphere is proportional to the cube of its radius, r cm.

m = 54 when r = 3.

(a) Find m when r = 5. (2)

m = ...

(b) Find r when m = 128. (2)

r = ...

Question 6

(2 marks)

y is directly proportional to x^3.

x is doubled.

By what number is y multiplied?

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $y = kx$, $28 = 4k$ — M1

$k = 7$, $y = 7x$ — A1

(b) $y = 7 \times 9 = 63$ — B1

Tip: find k first.

Question 2

$C = kL$, $9 = 6k$ — M1

$k = 1.5$ — M1

$C = 1.5 \times 14 = £21$ — A1

Tip: k is the cost of 1 m.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$y = kx^2$, $75 = 25k$ — M1

$k = 3$ — M1

$y = 3 \times 64 = 192$ — A1

Tip: square x before substituting.

Question 4

$y = k\sqrt{x}$, $12 = 3k$ — M1

$k = 4$ — M1

$y = 4 \times \sqrt{64} = 32$ — A1

Tip: the square root of 9 is 3.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$m = kr^3$, $54 = 27k$, $k = 2$ — M1

(a) $m = 2 \times 125 = 250$ — A1

(b) $128 = 2r^3$, $r^3 = 64$ — M1

$r = 4$ — A1

Tip: write the formula first.

Question 6

$y = k(2x)^3 = 8kx^3$ — M1

y is multiplied by 8 — A1

Tip: replace x with 2x.